Track zero sensor for Commodore 1541(-II)

In this document, two track zero sensors for drives used in the Commodore 1541 family of floppy drives are presented. The mechanisms involved are the ALPS drive employed in all variants except the 1541-II, and the Chinon drive often found in the 1541-II. The purpose of the sensor is to get rid of the notorious bumping these drives exhibit when trying to establish the outermost track position of the head. Adding the functionality involves expanding the mechanism with the sensor, and changing the ROM software to support it.

Identifying the mechanism model

Identifying the ALPS mechanism is easy; it is the only one using the push down mechanism to lock the disk in place. These mechanisms were employed in all variants of the 1541, except the 1541-II. Variants with black, brown and off-white fronts exist.

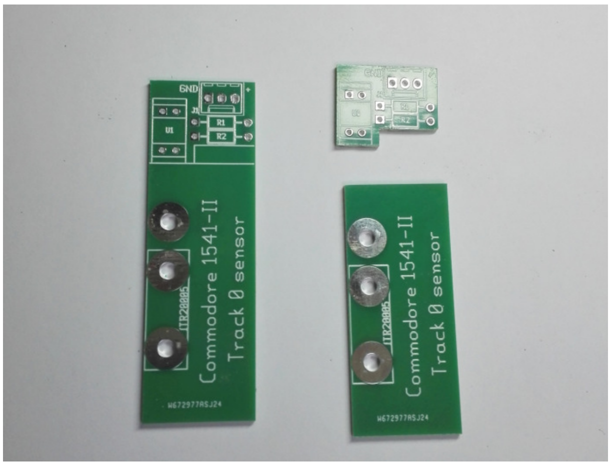
The Chinon mechanism was used in the 1541-II only and features a turn handle to lock the floppy. To properly identify it, you have to open the drive, and look for the identification Chinon on the mechanism.

Building the sensor (ALPS)

Building the sensor (Chinon)

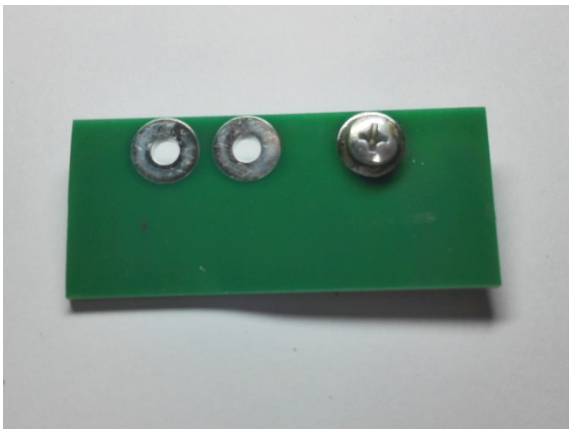
To build the sensor part, follow these steps:

Cut the PCB in two parts as shown using a saw or Dremel tool. Also remove the bottom part indicated by the silk lines of the smaller part using a file, Dremel tool and/or saw.



For the next step you need two M3x6 bolts, an M3 nut and the base plate PCB. The nut and bolts need to be solderable; stainless steel ones are not. The author used zinc plated hardware, brass should also work.

Insert a bolt through the hole for the optical sensor closest to the board edge, with the head of the bolt on the side without text. Secure the bolt with the nut. Solder the head of the bolt to the tin plated area surrounding it.

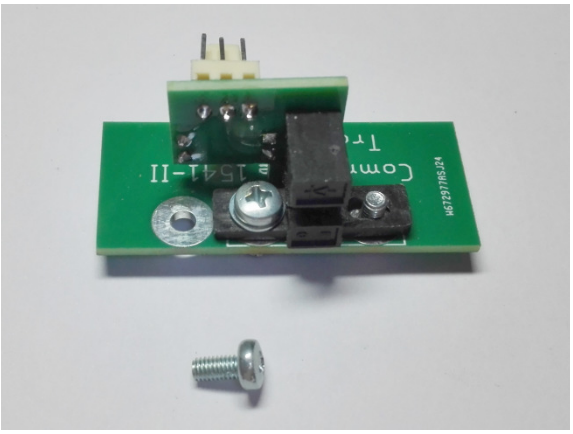


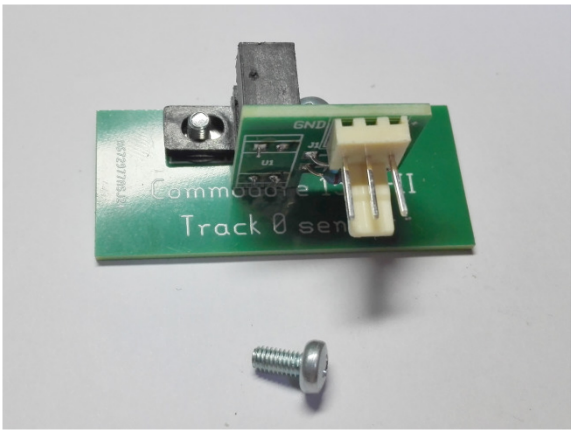
The protruding threaded part of the bolt will serve as a guide pin for the photo sensor.

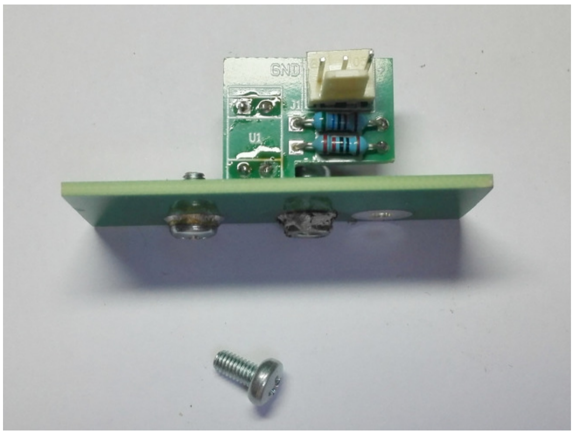
Remove the nut (after the lot has cooled down). Insert a second M3x6 bolt in the other hole for the optical sensor, this time the head is on the text side of the base plate. Secure the bolt with the nut and solder the nut to the tin plated area surrounding it.

Remove the bolt (after cooling down). Preparation for the base plate is now complete.

Build up the small PCB. The optical sensor is mounted on the non-text side of the PCB and soldered on the text side. The two resistors and the connector are located on the other side and soldered on the non-text side (see photos below). Attach the assembled smaller PCB to the base plate using a M3x6 bolt and an M3 small washer. The whole thing should look like this:







As can be seen, the thread part of the first bolt soldered is used as a guide pin and the sensor can be shifted sideways by loosening the second bolt, setting its position and tightening the second bolt again. The remaining hole in the base plate and the third bolt shown are used to mount the sensor assembly in the floppy mechanism.

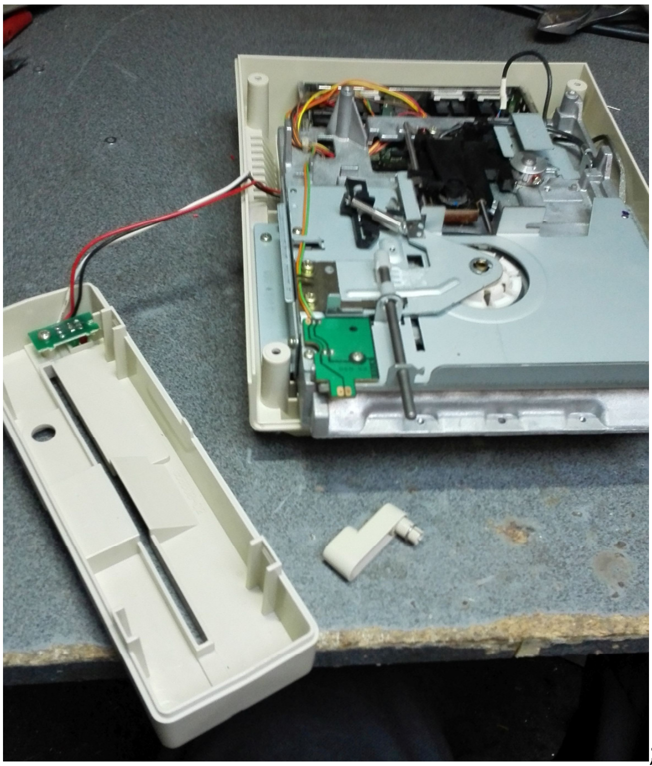
After checking everything fits and slides smoothly, remove the smaller PCB from the base plate.

Mounting the sensor in the drive (Chinon)

Take the 1541-II apart in the following order:

- Remove the four self tapping screws that attach the top cover to the bottom of the case;

- Remove the top cover and lay it aside;
- Remove the disk locking handle by pulling. Lay the handle aside;
- The front can now be taken off by unhooking the tabs at the bottom. Although not strictly required, the front can be separated from the rest by removing the self tapping screw that attaches the PCB holding the LEDs to the front. Move the front away or lay it aside. You should now have this: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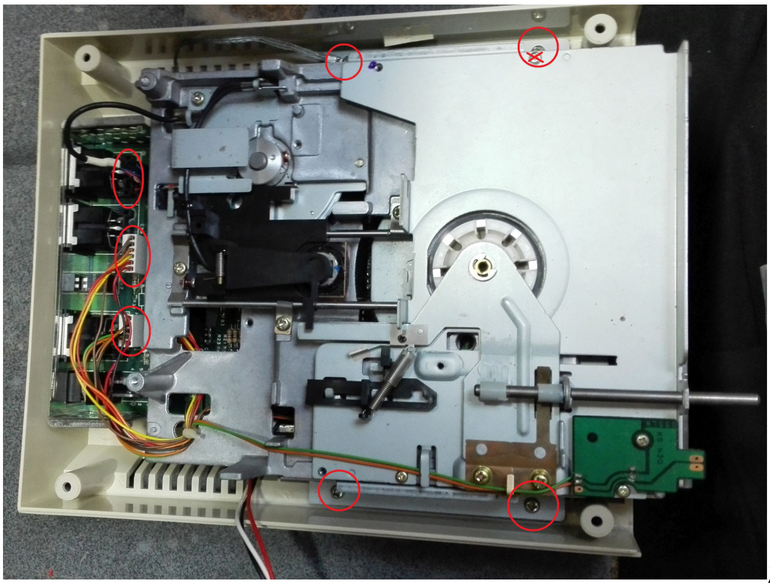
- The drive mechanism is connected to the main PCB with three connectors at the rear. Remove the three connectors from the main PCB. The connectors involved are indicated with red oval circles in the photograph below:
;
- The drive mechanism is mounted to the bottom part of the case with four self tapping screws indicated with red circles in the photograph above. Loosen these and lay them aside. Note that the screw at the right rear also connects a ground strap. The drive mechanism can now be taken out;
- For the moment lay the bottom case with the main PCB aside;

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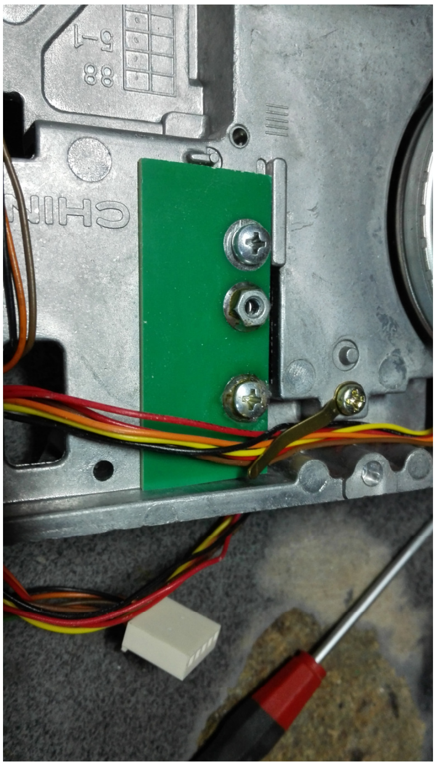
- Put the drive mechanism upside down with the rear facing you. Mount the base plate of the sensor using the third M3x6 bolt, using the pre-tapped hole already present in the mechanism's chassis:
  
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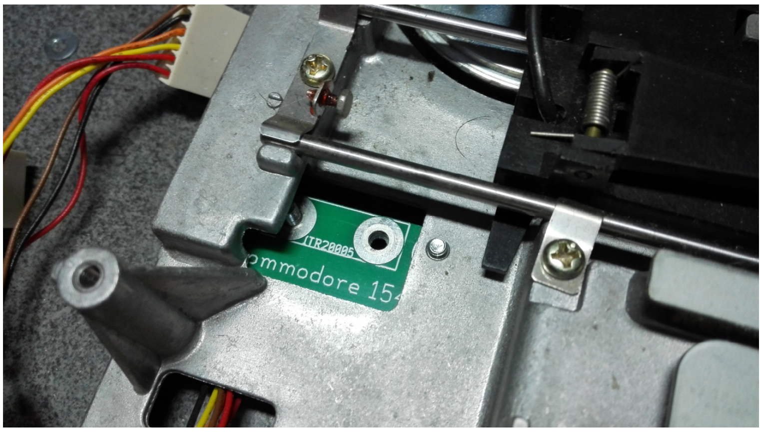
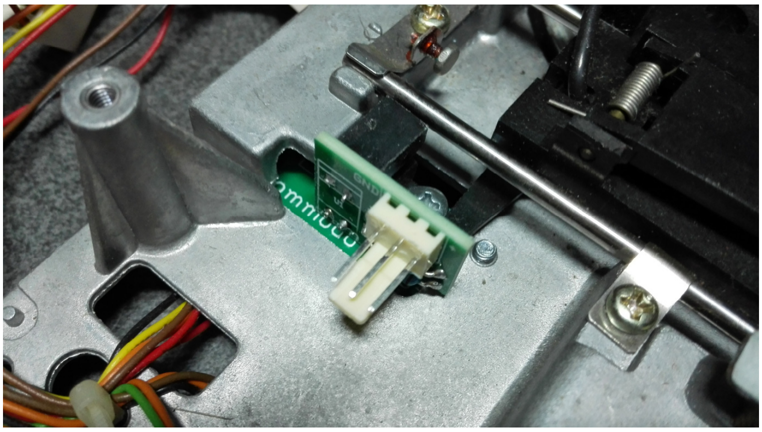
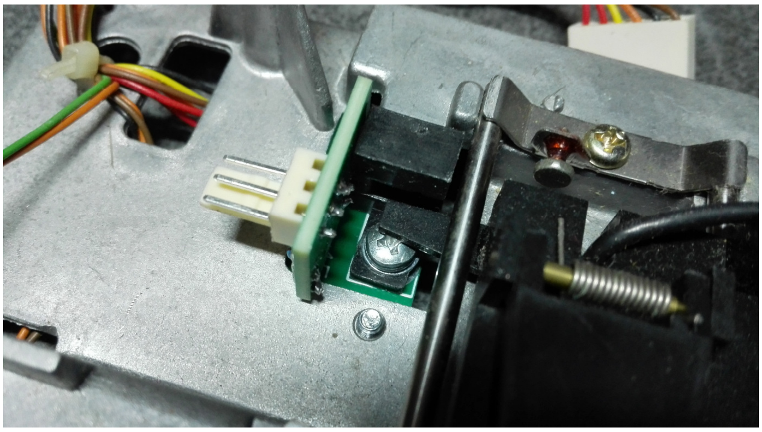
- Turn the drive mechanism over. The top side will look like this:

  Mount the sensor in place. First hook the rear hole in the optical sensor over the rear bolt on the base plate and then attach the optical sensor using an M3x6 bolt and the small washer. The end result should look like this:

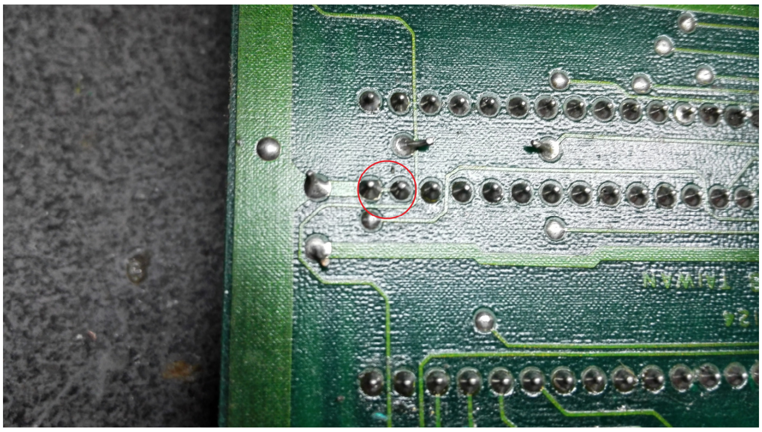
  Carefully move the head to the rear of the mechanism and check the vane for the sensor does not touch it;
- The drive mechanism is now ready for wiring and adjustment.

Changing the main PCB (1541-II Chinon)

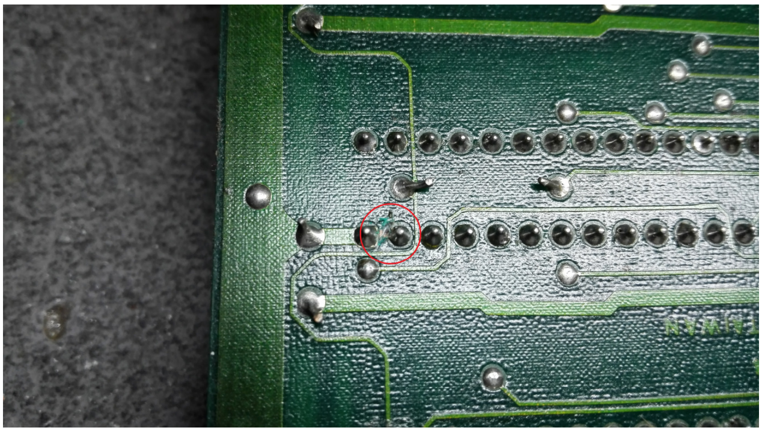
- The main PCB is in the bottom part of the case:

  Take the PBC out after removing the three self tapping screw holding it. The is one screw at the front and two at the back. The screw at the right rear holds the other end of the ground strap mentioned before;
- Turn the PCB over (solder side up) locate pin 1 of the front VIA. It is mark as a square island, as opposed to the other pins. It is connected to pin 2 of the VIA with a small copper trace:

- Cut the trace between pins 1 and 2 with a sharp object. Take care not the damage the solder islands. The result should look like this:



- Solder a wire to pin 2 of the VIA. In the following text the wire will be referred to as the brown wire. It carries the track 0 signal and will be connected to the middle connector pin of the sensor assembly: